A11.1 Principles of Collision Theory

Rates of reaction and energy changes

T-LEVELS

Learning outcomes

- Explain why reacting particles must collide.
- Explain activation energy as the minimum collision energy needed for bonds to break and reform.
- Distinguish successful and unsuccessful collisions using collision energy.
- Explain why particles must collide in the correct spatial orientation.
- Apply both requirements to unfamiliar particle-collision scenarios.

Starter: will every collision react?

- Two reactant particles collide. Must a reaction always occur?
- What must happen to existing bonds during many reactions?
- What might prevent a collision from producing products?
- Predict two conditions a collision may need to satisfy to be successful.

Starter answers

- No. Many collisions are unsuccessful.
- Existing bonds may need to break before atoms can rearrange and new bonds form.
- A collision may have insufficient energy or an unsuitable spatial orientation.
- Successful collisions require sufficient energy and the correct orientation.

Collision theory: the central idea

- For a reaction to occur, reactant particles must collide.
- However, collision alone is not sufficient.
- A successful collision must satisfy the energy requirement and the orientation requirement.
- If either requirement is not met, the collision is unsuccessful and products do not form.

A11.1.1 Activation energy

- Activation energy is the minimum energy that colliding particles must have for a successful reaction.
- Particles need enough collision energy for the required bonds to break and atoms to rearrange.
- New bonds can then form as products are produced.
- A collision with energy below the activation energy is unsuccessful.

Model: collision energy below activation energy

- Reactant particles collide.
- The collision energy is less than the activation energy.
- The required bond breaking/rearrangement cannot occur.
- The particles separate without forming the specified products.

Model: collision energy meets activation energy

- Reactant particles collide with energy equal to or greater than the activation energy.
- The collision has sufficient energy for the required bond changes.
- If the orientation is also correct, the collision can be successful.
- Products can then form.

Important distinction: energy is necessary, not sufficient

- A collision can have enough energy and still fail.
- The reacting parts of the particles must also meet in a suitable orientation.
- Collision theory therefore uses two tests: sufficient energy AND correct spatial orientation.
- This prevents the misconception that every high-energy collision must react.

Activity: Successful Collision Model

Teacher action

- Model collisions with energies below and above a stated activation energy.
- Require students to test energy first, then orientation.

Student action

- Classify each collision as potentially successful or unsuccessful.
- For each, compare collision energy with activation energy.
- Explain whether the collision has sufficient energy for bonds to break and reform.

Successful Collision Model: feedback

- Collision energy < activation energy → unsuccessful due to insufficient energy.
- Collision energy ≥ activation energy → energy requirement satisfied, but orientation must still be checked.
- A complete explanation refers to sufficient energy for the required bond breaking/rearrangement and formation of new bonds.

A11.1.2 Spatial orientation

- Reacting particles have a three-dimensional arrangement.
- For some collisions, particular atoms or reactive regions must meet one another.
- If the particles approach in an unsuitable orientation, the necessary bond changes cannot occur.
- Correct spatial orientation is therefore a second requirement for a successful collision.

Model: correct and incorrect orientation

- Incorrect orientation: particles collide, but the required reactive regions do not meet appropriately → no reaction.
- Correct orientation: the relevant parts of the reactants meet → the required bond rearrangement can occur if collision energy is sufficient.
- Orientation concerns how particles meet, not simply whether they touch.

Combining the two conditions

- Test 1 — Energy: is collision energy at least the activation energy?
- Test 2 — Orientation: are the particles aligned so the required atoms/regions can interact?
- Both yes → successful collision is possible.
- Either no → unsuccessful collision.

Activity: Orientation Challenge

Teacher action

- Show several particle-collision diagrams with different approaches.
- Model how to justify one correct and one incorrect orientation.

Student action

- Identify which collisions have the correct spatial orientation.
- Explain what is wrong with unsuccessful orientations.
- Then decide whether each collision would react when energy information is also supplied.

Guided application

- Reaction X has an activation energy of 45 kJ mol^{-1}.
- Collision A: 30 kJ mol^{-1}, correct orientation.
- Collision B: 55 kJ mol^{-1}, incorrect orientation.
- Collision C: 50 kJ mol^{-1}, correct orientation.
- For each collision, decide whether products can form and justify using both collision-theory conditions.

Guided application: answers

- A: unsuccessful — 30 kJ mol^{-1} is below the 45 kJ mol^{-1} activation energy, despite correct orientation.
- B: unsuccessful — energy is sufficient, but spatial orientation is incorrect.
- C: successful collision is possible — energy exceeds activation energy and orientation is correct.

Independent: Collision Energy Challenge

Teacher action

- Set the activation-energy threshold and clarify that students must justify, not just classify.
- Prompt use of the terms activation energy, collision energy and spatial orientation.

Student action

- Analyse unfamiliar collision scenarios independently.
- Determine whether each collision has sufficient energy.
- Check orientation and give a final successful/unsuccessful judgement with justification.

Common misconceptions

- ‘Particles react whenever they collide.’ Incorrect: energy and orientation conditions must be satisfied.
- ‘Activation energy is energy released by a reaction.’ Incorrect: it is the minimum energy required for a successful collision.
- ‘Enough energy guarantees reaction.’ Incorrect: spatial orientation must also be correct.
- ‘Correct orientation compensates for low energy.’ Incorrect: both conditions are required.

Check understanding

- 1. Define activation energy.
- 2. Explain why a collision below the activation energy is unsuccessful.
- 3. Explain what is meant by correct spatial orientation.
- 4. A collision has sufficient energy but incorrect orientation. Predict the outcome.
- 5. State the two requirements for a successful collision.

Check understanding: answers

- Activation energy: minimum energy colliding particles must have for a successful reaction.
- Below this threshold there is insufficient energy for the required bond breaking/rearrangement.
- Correct orientation means the particles meet so the relevant atoms/reactive regions are positioned for the required bond changes.
- No products form from that collision because the orientation requirement is not satisfied.
- Sufficient collision energy (at least activation energy) and correct spatial orientation.

Exam style application

- A reaction occurs when particles P and Q collide. The activation energy is 62 kJ mol^{-1}.
- (a) Collision 1 has 58 kJ mol^{-1} and correct orientation. Explain why it is unsuccessful. [2]
- (b) Collision 2 has 70 kJ mol^{-1} but incorrect orientation. Explain why it is unsuccessful. [2]
- (c) State the conditions needed for a collision between P and Q to form products. [2]
- (d) Explain why collision theory predicts that many collisions do not produce a reaction. [2]

Exam style application: answers

- (a) 58 kJ mol^{-1} is below the 62 kJ mol^{-1} activation energy; there is insufficient energy for the required bond changes.
- (b) Energy exceeds activation energy, but the particles are not in the correct spatial orientation for the required interaction/bond rearrangement.
- (c) Collision energy must be at least the activation energy AND the particles must have the correct spatial orientation.
- (d) Many collisions fail one or both requirements: insufficient energy and/or incorrect orientation.

Plenary: complete the model

- Reactant particles must _____.
- The collision must have energy ≥ _____.
- This energy is needed so bonds can _____ and new bonds can _____.
- The particles must also have the correct spatial _____.
- Only collisions satisfying both requirements are _____ collisions.

Plenary answers

- collide
- activation energy
- break / reform (or rearrange) and form
- orientation
- successful